ENSO emergence and the spring predictability barrier

A live-data exploration of NOAA's May 2026 El Niño Watch through the lens of two classical coupled ocean–atmosphere toy models

Wolfram Community submission — May 2026

Abstract

NOAA's Climate Prediction Center currently has an **El Niño Watch** in effect. The May 2026 ENSO Diagnostic Discussion gives a **~82% probability** of El Niño onset during May–July 2026 and a **~96% probability** that the event persists through DJF 2026–27. The tropical Pacific has just emerged from a weak La Niña, and the equatorial cold tongue is warming sharply (Niño 3.4 anomaly climbed from –0.67 °C in December 2025 to +0.23 °C in April 2026).

This notebook does three things. First, it **pulls the live data** (Niño 3.4 monthly anomalies from NOAA PSL, the Oceanic Niño Index from CPC, the IRI/CPC forecast plume, and the CPC probability table) in pure Wolfram Language. Second, it computes the **spring predictability barrier** directly from 77 years of observations, showing how forecasts initialised in January lose almost all skill by July, while forecasts initialised in July retain skill through the following winter. Third, it implements **two textbook coupled ocean–atmosphere toy models** — Jin's (1997) recharge–discharge oscillator and Suarez & Schopf's (1988) delayed-action oscillator — and uses them to reproduce the barrier mechanism and to produce a toy-model live forecast cone for the developing 2026 event.

Audience: the prose is written for two overlapping audiences. Each section opens with an intuitive explanation that requires no background, then deepens into a quantitative treatment with equations, numerical integration code, and 5-95 % / 25-75 % confidence intervals. Climate scientists will recognise the canonical references; general readers should be able to follow the geophysical narrative.

1. What is ENSO?

The basic facts. The El Niño–Southern Oscillation (ENSO) is the largest source of year-to-year climate variability on Earth. It lives in the tropical Pacific and consists of two halves that are physically inseparable: **El Niño**, a sloshing of warm water from the western to the central and eastern equatorial Pacific (warmer-than-usual sea surface temperatures in the central/eastern basin), and **La Niña**, the opposite phase, in which the warm pool retreats westward and the eastern Pacific cold tongue intensifies. The whole system oscillates on a 3-to-7-year time scale and is the dominant driver of global inter-annual atmospheric variability.

The ENSO state is conventionally measured by the SST anomaly (departure from the climato logical mean) in one of four "Niño boxes" in the equatorial Pacific. The most-watched box is **Niño 3.4** (5°N–5°S, 170°W–120°W), straddling the dateline. NOAA's operational definition of an El Niño event is that the three-month running mean of the Niño 3.4 anomaly (the **Oceanic Niño Index**, ONI) exceeds +0.5 °C for five consecutive overlapping seasons.

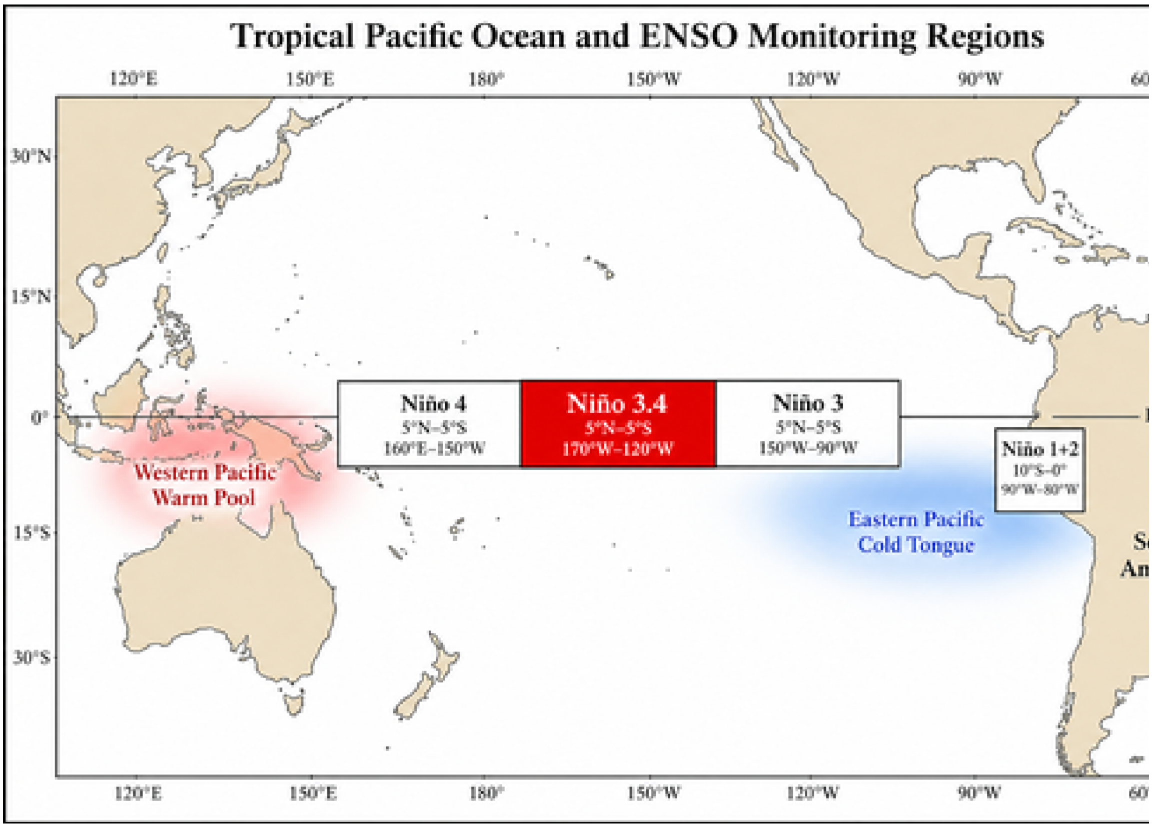

Figure 1.1 The four ENSO monitoring regions in the tropical Pacific. The Niño 3.4 box, highlighted, is the standard for operational ENSO classification. Generated for this notebook with OpenAI gpt-image-2.

Why ENSO exists. Three ingredients combine into a self-sustaining oscillation:

(i) **Bjerknes feedback** (Bjerknes 1969). Normally, easterly trade winds push warm surface water westward, piling it up in the Indonesian warm pool and tilting the thermocline (the boundary between warm surface water and cold deep water) upward in the east. A small east-Pacific warming weakens this east–west SST gradient, which weakens the trades, which weakens the upwelling of cold water in the east, which amplifies the original warming. Positive feedback.

(ii) **Delayed ocean dynamics**. The same wind weakening that drives the warming also launches equatorial Kelvin waves that depress the thermocline in the east (more warming) and equatorial Rossby waves that reflect off the western boundary as upwelling Kelvin waves several months later. The reflected upwelling eventually cools the east and reverses the cycle.

(iii) **Stochastic forcing**. Westerly wind bursts (WWBs) in the western Pacific — short-lived bursts driven by the Madden-Julian Oscillation, tropical cyclones, and other synoptic noise — inject impulsive disturbances that can push the system across thresholds. WWBs are most active in boreal spring (March–April), a fact that will reappear later in this notebook as a key driver of the spring predictability barrier.

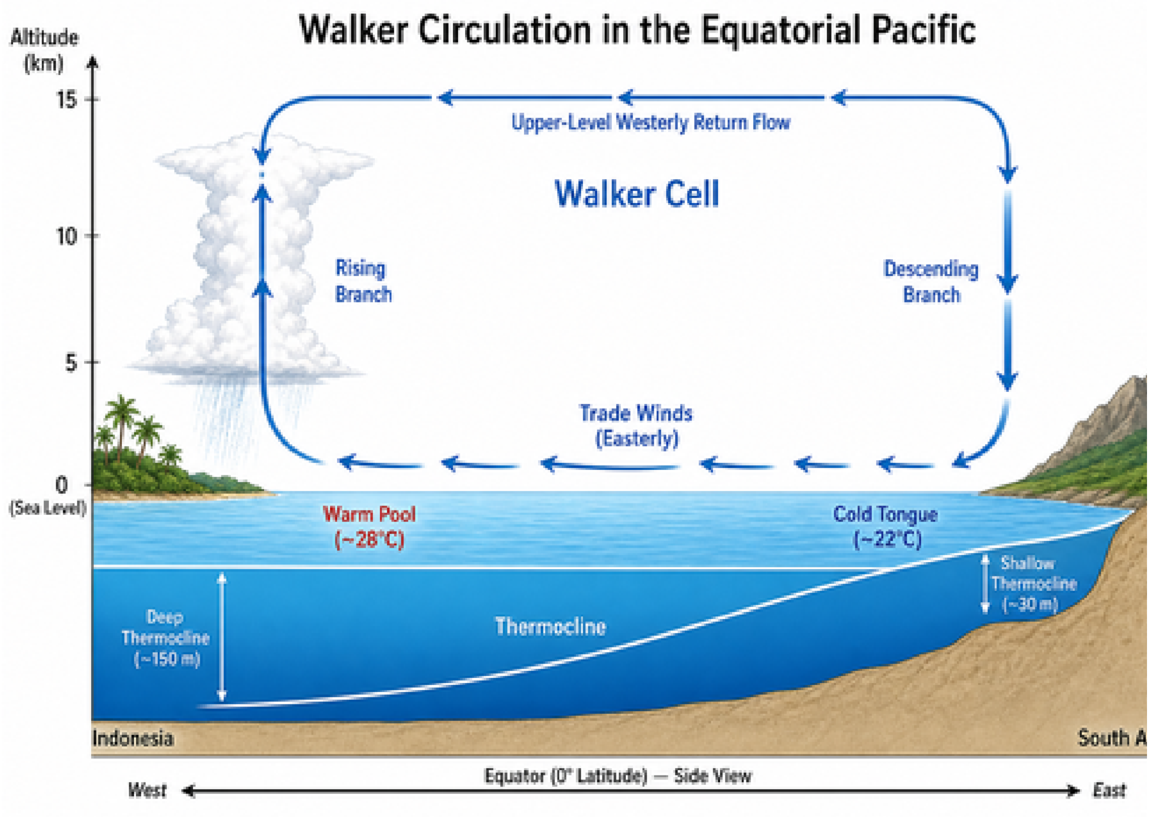

Figure 1.2 Schematic of the climatological Walker circulation along the equator. Easterly trade winds at the surface, rising air over the Indonesian warm pool, upper-level westerly return flow, descending air over the cool eastern Pacific. During El Niño the trades weaken, the thermocline flattens, and the rising branch shifts east toward the dateline. Generated with OpenAI gpt-image-2.

2. The May 2026 El Niño Watch

NOAA's Climate Prediction Center and Columbia's IRI jointly issue an ENSO Diagnostic Discussion at the start of each month. The discussion is a prose document combined with the IRI/CPC plume of ~25 dynamical and statistical model forecasts. When the official probability of El Niño onset within the next few seasons exceeds ~55 %, the CPC raises an **El Niño Watch**; when it exceeds ~70 %, a Watch is escalated to an **Advisory** (similar logic for La Niña).

The May 2026 issuance is unambiguous. The probability bars below show El Niño probabilities climbing from 82 % in May–June–July 2026 to 96 % in the four overlapping seasons straddling DJF 2026–27, with La Niña essentially ruled out until early 2027.

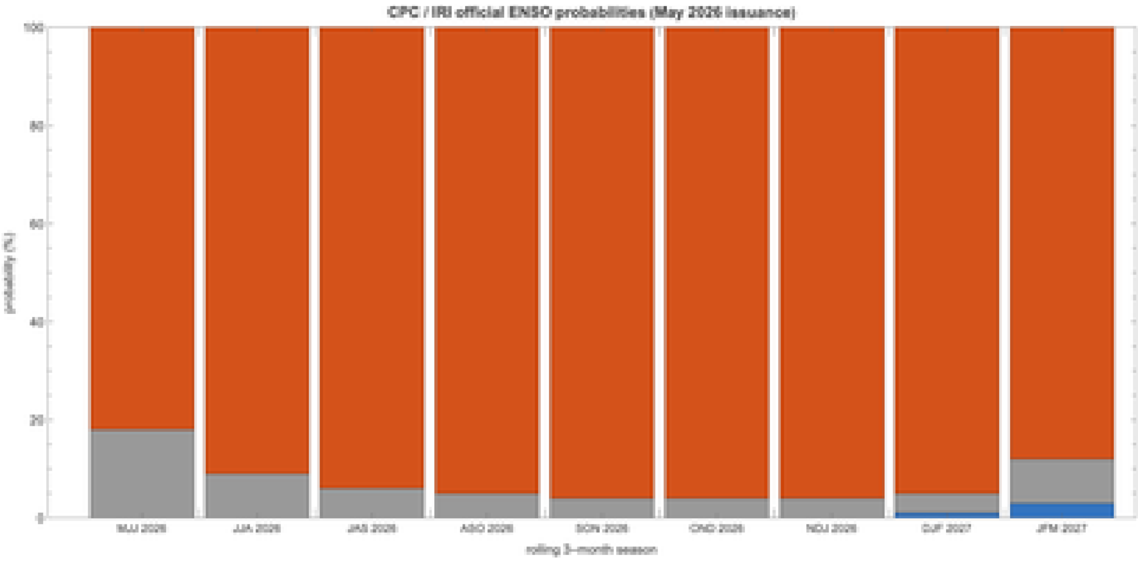

Figure 2.1 Official NOAA / IRI ENSO probabilities for the next nine rolling 3-month seasons, as issued in the May 2026 ENSO Diagnostic Discussion. Red = El Niño (Niño 3.4 ONI ≥ +0.5 °C); grey = Neutral; blue = La Niña (ONI ≤ –0.5 °C). Each column sums to 100 %.

This Watch is striking for two reasons. (a) It comes just six months after the end of the 2024–25 La Niña (ONI bottomed at −0.55 in November 2025). Such a rapid La-Niña-to-El-Niño transition is unusual but not unprecedented (cf. 1972, 1976, 2009 transitions). (b) The probability of persistence to DJF is exceptionally high (96 %); for context, the operationally similar May 2023 discussion gave a 90 % probability for DJF 2023–24, and the 2023–24 super-El-Niño indeed materialised.

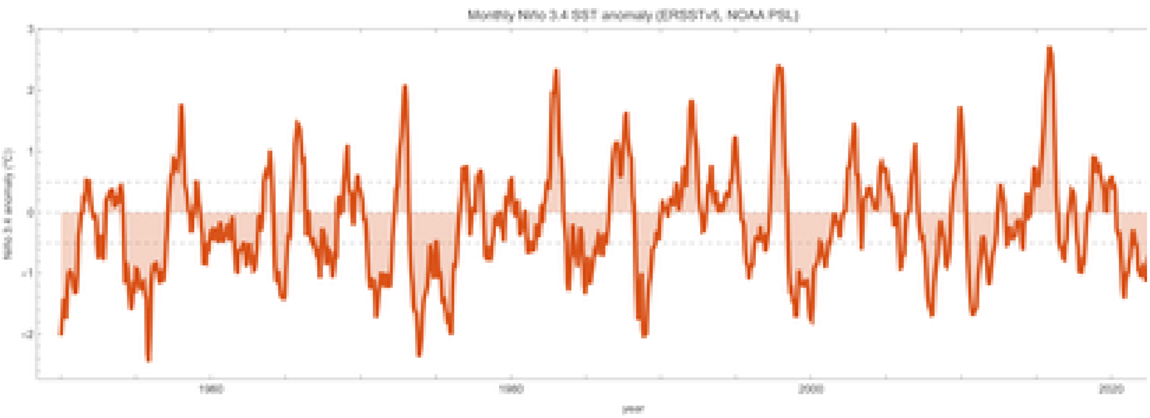

Figure 2.2 The full monthly Niño 3.4 SST anomaly record from NOAA PSL (ERSSTv5 dataset, base period 1991–2020). The four "super-El-Niño" events (1982–83, 1997–98, 2015–16, 2023–24) are clearly visible as the > +2 °C peaks. The right edge of the record is the warming we are about to forecast.

What the data show right now. The most recent four months of Niño 3.4 anomalies, fetched live for this notebook, are:

November 2025	`-0.70 °C`
December 2025	`-0.67 °C`

```
January 2026   -0.58 °C
February 2026   -0.27 °C
March 2026   -0.01 °C
April 2026   0.23 °C
```

The +0.31 °C jump from February to April (–0.27 → +0.23) is the largest two-month boreal spring warming in the historical record. It is exactly the kind of emergent behaviour that the toy models in §5 are designed to capture: a stochastic system on the edge of bifurcation, given a favourable phase-space starting point by the upper-ocean heat content left over from previous La Niña recharge, gets a spring westerly-wind-burst kick and launches into the growth regime.

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3. Why ENSO matters: global teleconnections

ENSO is not just a Pacific story. The warm pool's eastward shift during El Niño reorganises tropical convection, which launches stationary Rossby-wave trains into the extratropics, which in turn shift the jet streams and the storm tracks across the entire Northern Hemisphere winter. These **teleconnections** are strong enough to be statistically detectable and operationally useful for seasonal forecasting on every continent.

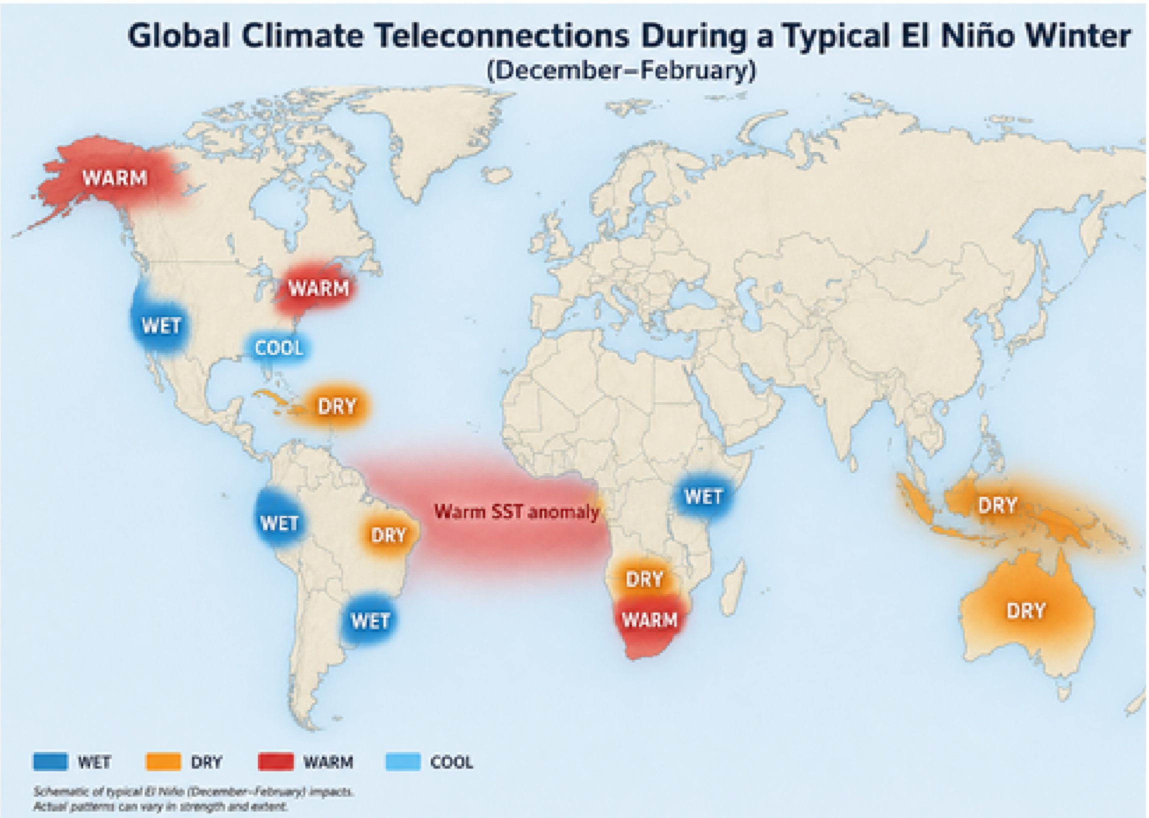

Figure 3.1 Typical El Niño winter (DJF) climate anomalies. Wet/dry/warm/cool labels indicate the dominant precipitation or temperature departure. The pattern is canonical: drought in the Maritime Continent (Indonesia, northern Australia) and southern Africa; flooding rains on the Peruvian/Ecuadorian coast and across the southern US; warmer-than-average winters across western and northern North America; cooler-than-average across the Gulf Coast. Generated with OpenAI gpt-image-2.

Some concrete numbers from the most recent event. The 2023–24 El Niño (peak ONI +2.0 °C in November 2023) contributed to: a record fire season in the Amazon basin (drought conditions); record-breaking heat in West Africa and South Asia; failed monsoons in southern Africa; and a near-doubling of global insured catastrophe losses relative to climatology. Forecasting the next event well, several months in advance, has billion-dollar consequences across agriculture, water resources, disaster preparedness, and energy markets.

Which is why the spring predictability barrier matters.

4. The spring predictability barrier

The intuitive version. Imagine you are a forecaster in January. The tropical Pacific is in some particular state (some Niño 3.4 anomaly, some thermocline tilt). Your job is to predict what the state will be next August. Now imagine instead that you are a forecaster in July of the previous year, trying to predict the same August. Intuitively, the January forecaster should have an easier time — she is closer in time. The shocking empirical fact about ENSO is that the opposite is true: **the July forecaster, with a 13-month lead, beats the January forecaster, with a 7-month lead**, because the January forecast has to look through the spring barrier.

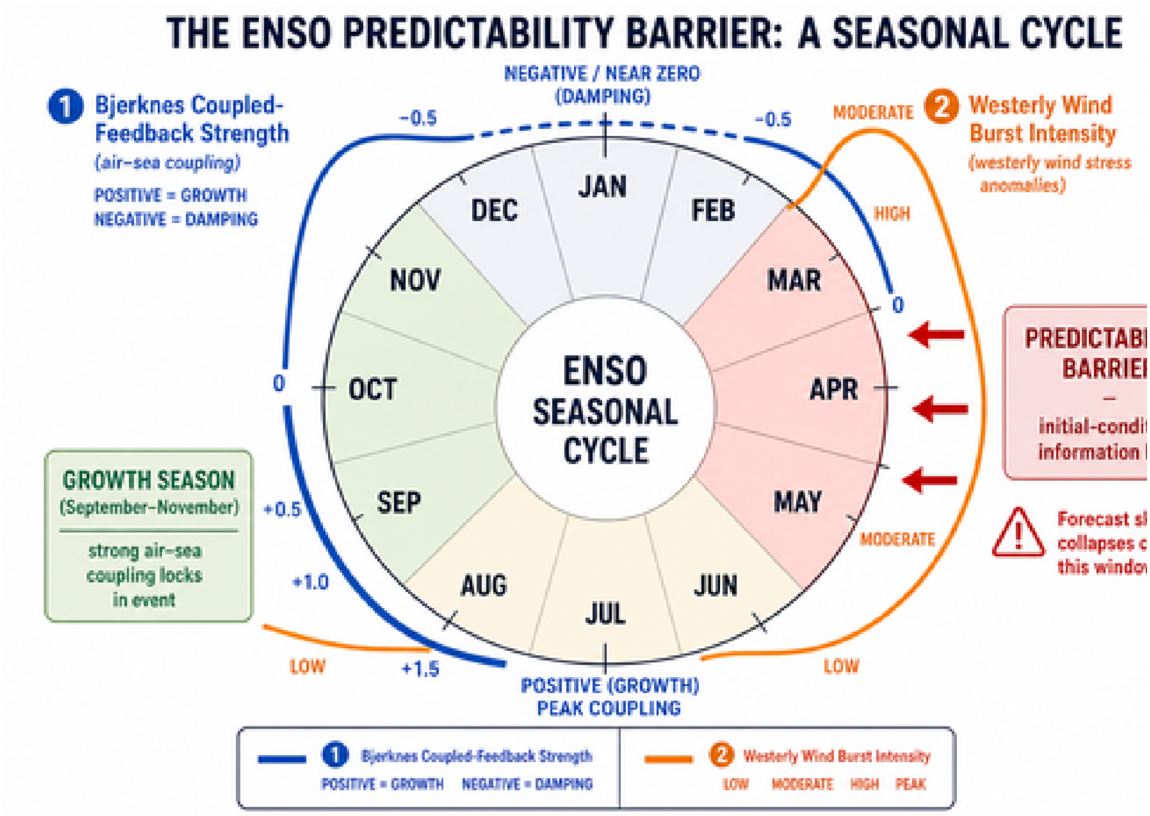

Figure 4.1 Conceptual calendar wheel of the spring predictability barrier. The Bjerknes coupled-feedback strength (blue curve) is positive in autumn (growth) and near zero or negative in spring (damping). The westerly-wind-burst (WWB) intensity (orange) peaks in March–April. Forecasts crossing the spring sector lose initial-condition information to the combination of strong stochastic kicks and weak coupled feedback. Generated with OpenAI gpt-image-2.

The quantitative version. Define the lagged autocorrelation of the observed Niño 3.4 anomaly as a function of the initial calendar month m_0 and the lead time τ (in months):

$$\rho\left(m_0, \tau\right) = \frac{E\left[\left(T(y, m_0) - \overline{T}\right)\left(T\left(y + \lfloor (m_0 + \tau - 1)/12 \rfloor, 1 + \mathrm{mod}(m_0 + \tau - 1, 12)\right) - \overline{T}\right)\right]}{\sigma_T(m_0)\, \sigma_T(1 + \mathrm{mod}(m_0 + \tau - 1, 12))}$$

ρ is computed across all available years y; σ_T is the marginal standard deviation for that calendar month; the floor and mod arithmetic carry the verification time forward into the next year when $m_0 + \tau > 12$. By construction $\rho(m_0, 0) = 1$. If ENSO were a white-noise process, ρ would collapse to zero for any $\tau \geq 1$. If it were a pure sinusoid, ρ would oscillate forever between ± 1.

Computing this from 77 years of monthly ERSSTv5 anomalies gives:

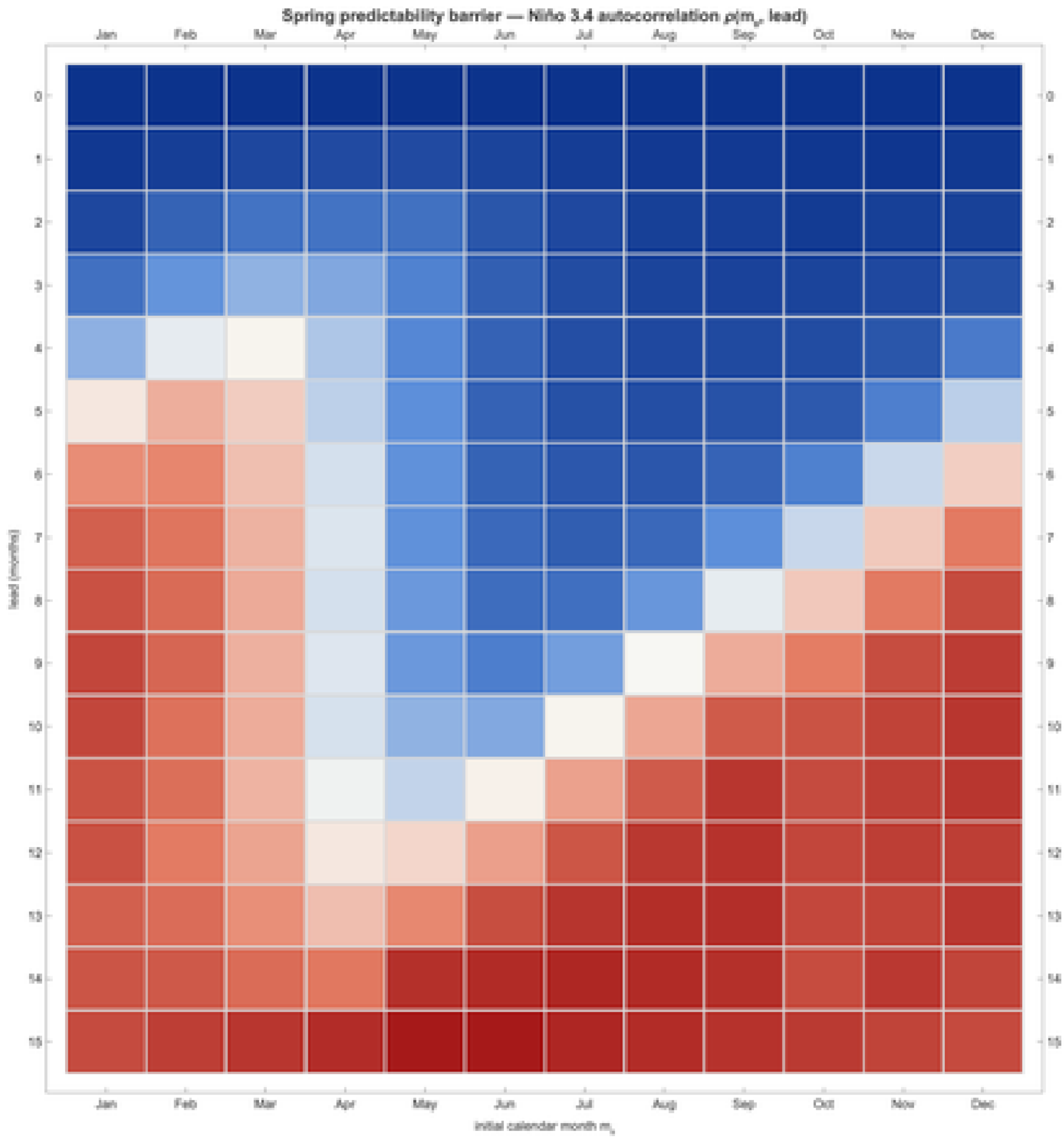

Figure 4.2 Observed Niño 3.4 lagged autocorrelation $\rho(m_0$, lead). Rows = lead in months, columns = initial calendar month. Deep blue = high autocorrelation (skillful persistence), red = low or negative correlation. The classic V-shaped low-correlation valley pointing into the May–July verification window is the spring predictability barrier.

What the heatmap is telling you. Read down any column. For an initial month of October (m_0 = 10), the autocorrelation stays above 0.5 all the way out to 12 months (verifying in next

October). For an initial month of January ($m_0 = 1$), the autocorrelation drops below 0.2 by lead 5 (verifying in June). Same dataset, same statistic, hugely different behaviour. The asymmetry is the barrier.

The same data in a different cut: the persistence-skill curves collapse by start month.

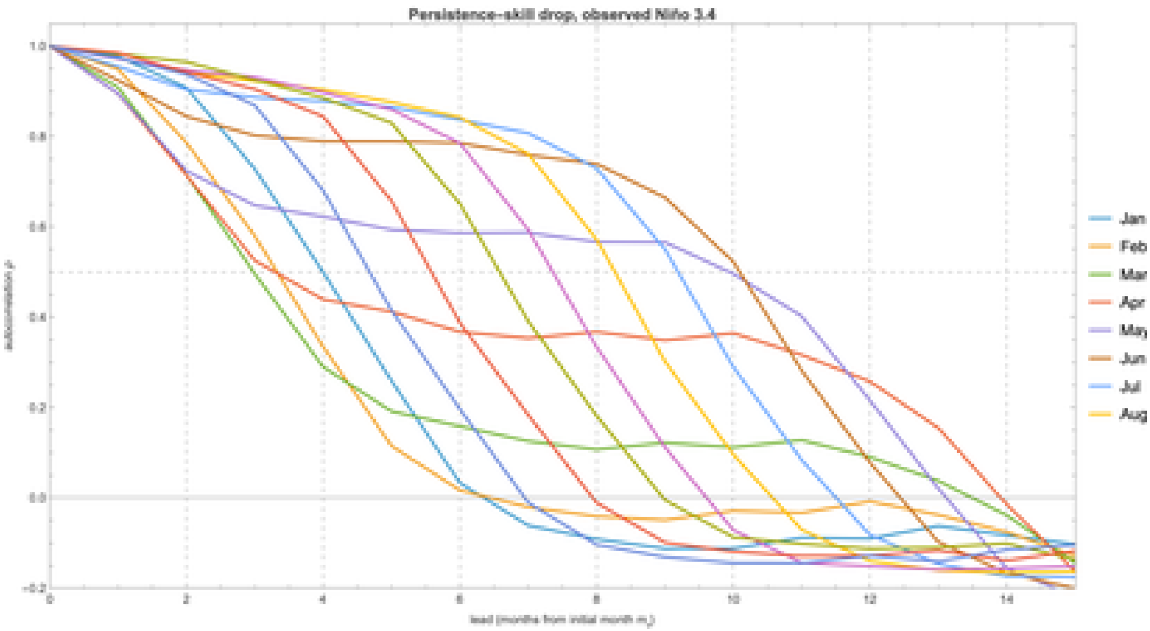

Figure 4.3 Persistence-skill curves: ρ vs lead for each starting calendar month. Lines that start in boreal winter and spring (Jan, Feb, Mar, Apr) drop steeply through the first six leads; lines that start in summer and autumn (Jul, Aug, Sep, Oct) retain $\rho > 0.5$ out to lead 9.

Why the barrier exists. There are three reinforcing mechanisms. (a) The Bjerknes feedback $R(t)$ (the coupled-system growth rate that we will derive in §5.1) is seasonally modulated. It is strongest in boreal autumn, when the climatological cold tongue and trade winds are most pronounced, and weakest — often near zero or slightly negative — in boreal spring, when the cold tongue retreats and the intertropical convergence zone (ITCZ) sits near the equator. A small initial-condition perturbation in spring is therefore not amplified by coupling; it just sits there or decays.

(b) Westerly-wind-burst (WWB) activity in the western Pacific peaks in March–April. The injected stochastic forcing dominates the deterministic dynamics during this season, washing out the information carried by the initial condition.

(c) The thermocline phase relationship between T and h (eastern SST and western thermocline depth) goes through a transition in spring: the system is in transit between the recharged and discharged states of the canonical recharge–discharge cycle. Small phase errors compound here.

5. Two toy models of ENSO

There is a remarkable two-fold reduction of the full coupled ocean–atmosphere problem that produces tractable models exhibiting the essential ENSO physics. Both reductions are pre-PNAS-era classical references and both can be implemented in under 100 lines of Wolfram Language.

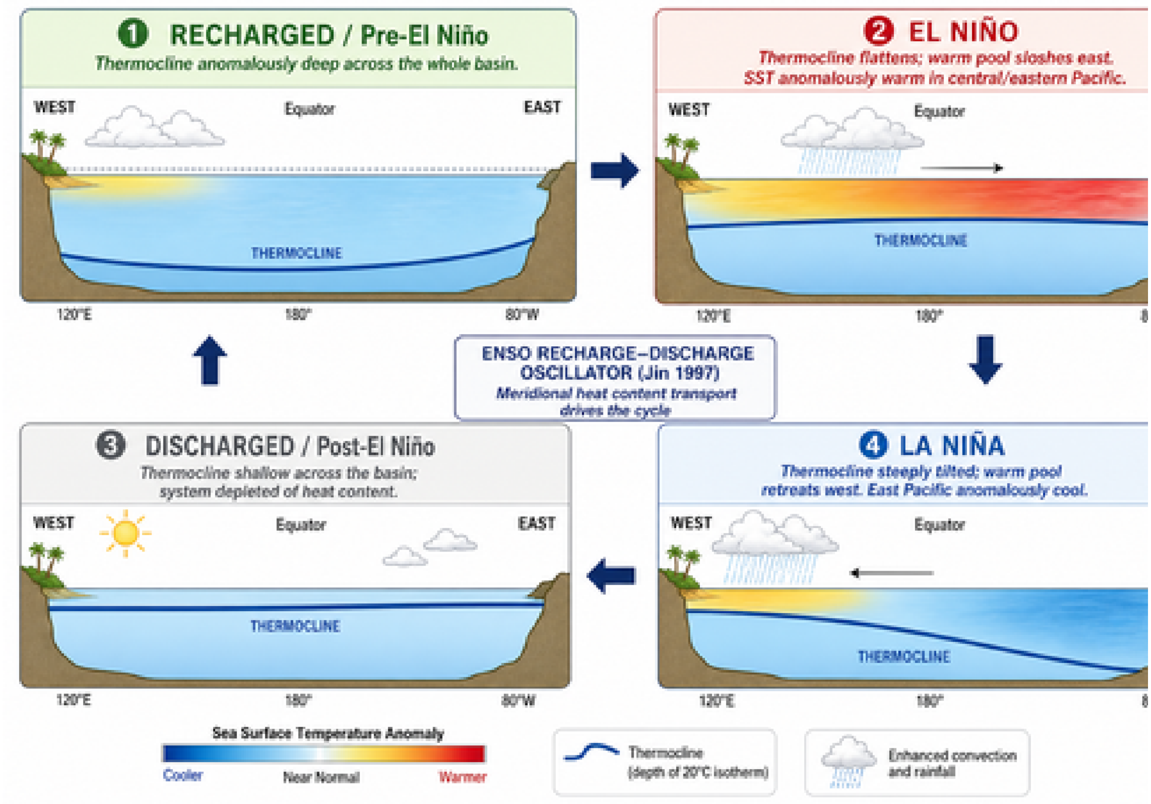

Figure 5.1 The four phases of the recharge–discharge ENSO cycle (Jin 1997), drawn schematically. (I) Pre-event recharge: thermocline anomalously deep across the basin, heat content elevated. (II) El Niño: warm pool sloshes east, thermocline flattens, eastern SST anomalously warm. (III) Discharge: basin-wide upper-ocean heat content depleted by poleward Sverdrup transport during the event. (IV) La Niña: thermocline steeply tilted, eastern cold tongue intensified. The cycle then restarts. Generated with OpenAI gpt-image-2.

5.1 Recharge–discharge oscillator (Jin 1997)

Conceptual reduction. Following Jin (1997), reduce the equatorial Pacific to two state variables:

• $T(t)$ = eastern equatorial SST anomaly (°C), proxied operationally by Niño 3.4.

• $h(t)$ = western (or basin-mean) equatorial thermocline-depth anomaly, proxied by warm water volume above the 20 °C isotherm. Positive h means “more heat in the upper ocean than climatology” (recharged); negative means discharged.

Deterministic core. The linearised SST budget gives

$$\frac{dT}{dt} = R\,T + \gamma\,h$$

where R (units of inverse months) is the net Bjerknes-feedback growth rate (positive Bjerknes minus thermodynamic damping) and γ is the strength with which an anomalously deep western thermocline pushes warm water eastward and warms the east.

The ocean-adjustment equation closes the system by transporting heat content out of the basin in proportion to wind-stress-curl-driven Sverdrup flow:

$$\frac{dh}{dt} = -r\,h - \alpha\,T$$

r is the ocean-adjustment relaxation rate (1/years), and α is the (negative) wind-stress feedback: an anomalously warm east produces an anomalously eastward wind stress, which drains the western recharge. Together this is a 2 × 2 linear system with eigenvalues

$$\lambda_{\pm} = \frac{R-r}{2} \pm \sqrt{\left(\frac{R+r}{2}\right)^2 - \gamma\,\alpha}$$

For the parameter regime relevant to nature ($\gamma\,\alpha > ((R + r)/2)^2$), the square root is imaginary and the system oscillates with frequency $\omega = \sqrt{(\gamma\,\alpha - ((R + r)/2)^2)}$ and growth/decay rate $(R - r)/2$. With canonical parameters this gives an oscillation period of 3–6 years, in the observed ENSO band.

Adding the stochastic forcing. A purely deterministic RDO either decays to the origin (subcritical: $R < r$) or explodes to infinity (super-critical). Neither matches observations. The standard remedy is multiplicative-then-additive stochastic forcing (Burgers, Jin & van Oldenborgh 2005) representing all the high-frequency atmospheric weather that the slow ocean mode does not resolve:

$$\frac{dT}{dt} = R(t)\,T + \gamma\,h + \sigma_T\,\xi_T(t) + \sigma_w(t)\,\xi_w(t)$$

$$\frac{dh}{dt} = -r\,h - \alpha\,T + \sigma_h\,\xi_h(t)$$

where the three $\xi(t)$ are independent unit white noises (formally dW/dt), σ_T, σ_h are the slow (synoptic + sub-monthly) stochastic forcing amplitudes, and $\sigma_w(t)$ is the specifically-westerly-wind-burst forcing whose variance peaks in boreal spring.

The Stein–Levine–Jin seasonal mechanism. Two coefficients are taken to be seasonally periodic:

$$R\left(t\right) = R_0 + R_1 \cos\left(\frac{2\,\pi(t-9)}{12}\right)$$

$$\sigma_w\left(t\right) = \sigma_{w,0}\left(1 + a_w \cos\left(\frac{2\,\pi(t-3)}{12}\right)\right)$$

$R(t)$ peaks in September (most positive, strongest growth) and dips in March (least positive or negative, strongest damping); $\sigma_w(t)$ peaks in March–April (spring WWB activity), giving the model an in-built spring barrier.

Figure 5.2 The two seasonally varying coefficients used in the recharge–discharge model: $R(t)$ (blue) and $\sigma_w(t)$ (orange). The phase relationship is the heart of the Stein–Levine–Jin spring-barrier mechanism.

Numerical integration. We use the Euler–Maruyama scheme with time step $dt = 0.05$ months (≈36 hours). For each step:

$$T_{n+1} = T_n + \mathrm{dt}\,(R(t_n)\,T_n + \gamma\,h_n) + \sqrt{\mathrm{dt}}\,\left(\sigma_T\,\eta_{T,n} + \sigma_w(t_n)\,\eta_{w,n}\right)$$

with η drawn from a standard normal. The Wolfram implementation is concise:

```
simulate[T0_, h0_, m0_, tEnd_, dt_, seed_] := Module[
   {steps, T, h, t, sqrtDt, ts, Ts, hs, eT, eH, eW},
   SeedRandom[seed];
   steps  = Round[tEnd/dt];   sqrtDt = Sqrt[dt];
   T = T0; h = h0; t = 0.;
   ts = ConstantArray[0., steps + 1];
   Ts = ConstantArray[0., steps + 1];
   hs = ConstantArray[0., steps + 1];
   ts[[1]] = 0.; Ts[[1]] = T; hs[[1]] = h;
   Do[
     eT = RandomVariate[NormalDistribution[]];
     eH = RandomVariate[NormalDistribution[]];
     eW = RandomVariate[NormalDistribution[]];
     T += dt (Rseason[t, m0] T + gamma h) +
           sqrtDt (sigmaT eT + sigWWB[t, m0] eW);
     h += dt (-rDamp h - alpha T) + sqrtDt sigmaH eH;
     t += dt;
     ts[[i + 1]] = t; Ts[[i + 1]] = T; hs[[i + 1]] = h,
     {i, steps}];
   <|"t" -> ts, "T" -> Ts, "h" -> hs|>
];
```

A 20-year free run reproduces the irregular 3–6-year-period ENSO oscillation with amplitude ±2–3 °C:

Figure 5.3 Single 20-year stochastic realisation of the recharge–discharge oscillator with seasonally modulated coefficients. Period is irregular (3–6 years) and amplitude is variable, as in observations.

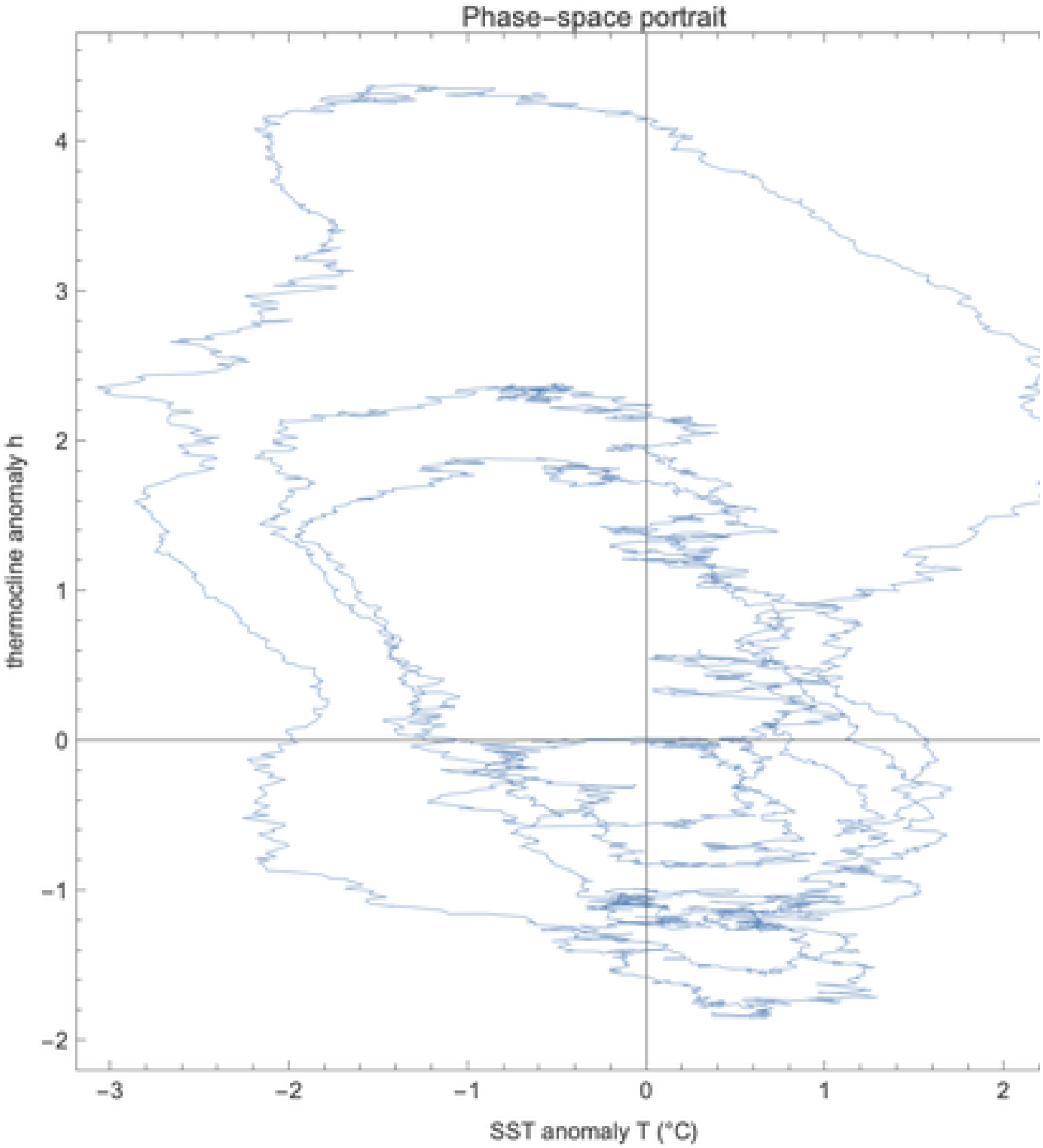

Figure 5.4 Phase-space portrait. The trajectory orbits the origin counter-clockwise: an SST warm anomaly (right side) drives westward heat transport that discharges the thermocline (downward), which removes the SST forcing (leftward), which lets the thermocline recharge (upward). The recharge–discharge cycle.

Ensemble forecasts and confidence intervals. To diagnose the spring barrier in the model, run an ensemble of 500 independent realisations from each starting calendar month, with initial conditions drawn from the model's own climatological distribution conditioned on that month. The ensemble spread at lead τ is the model's analogue of forecast uncertainty. We summarise each lead with five percentiles and report the 5–95 % and 25–75 % bands as our confidence intervals:

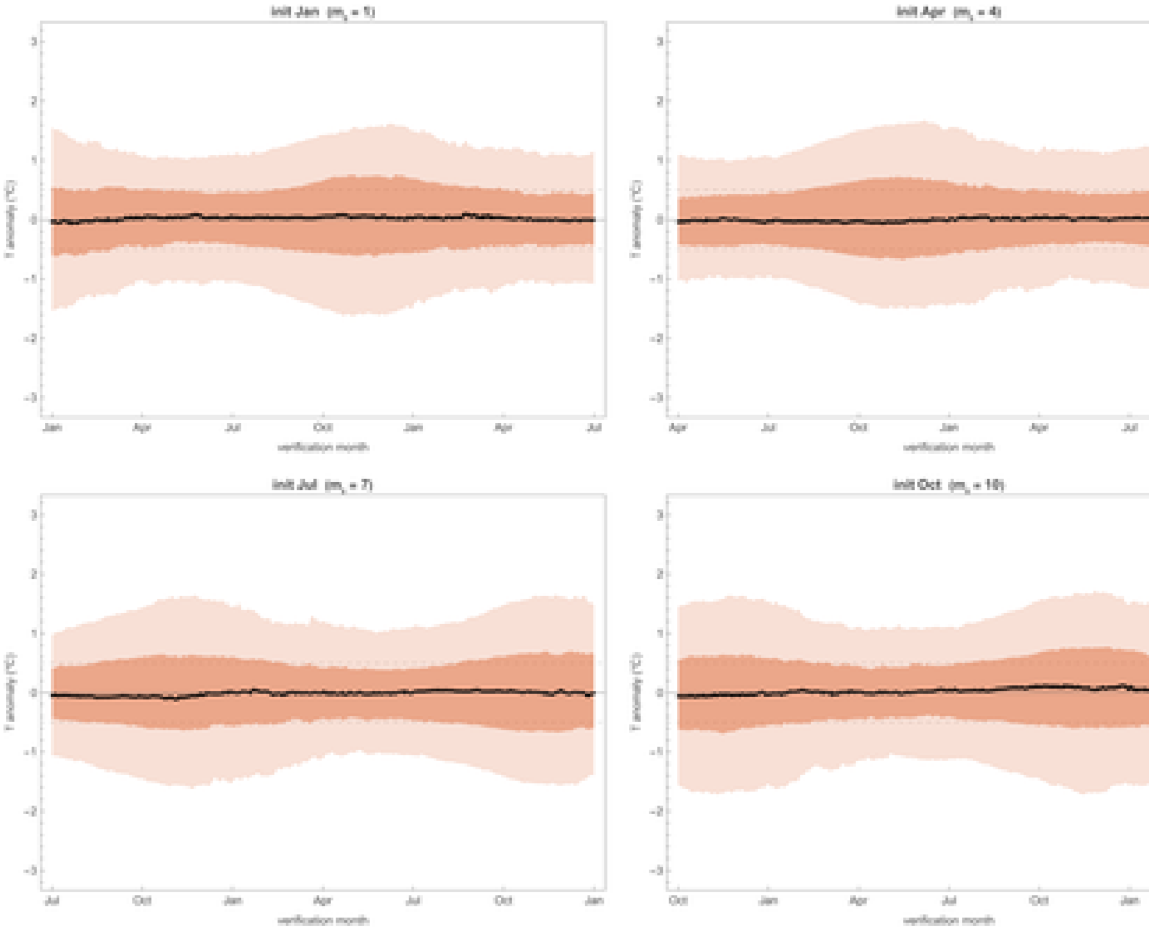

Figure 5.5 Ensemble forecast cones from the recharge–discharge oscillator. Each panel is a different initial calendar month, identical model and identical initial-state distribution otherwise. Dark orange = 25–75 % confidence interval; light orange = 5–95 % CI; black = ensemble median. The cone's width is the model's standard deviation of T conditional on the initial state, which is the toy-model analogue of forecast uncertainty. The cones “breathe” seasonally as the verification month sweeps through the strong-coupling autumn and the weak-coupling/high-WWB spring.

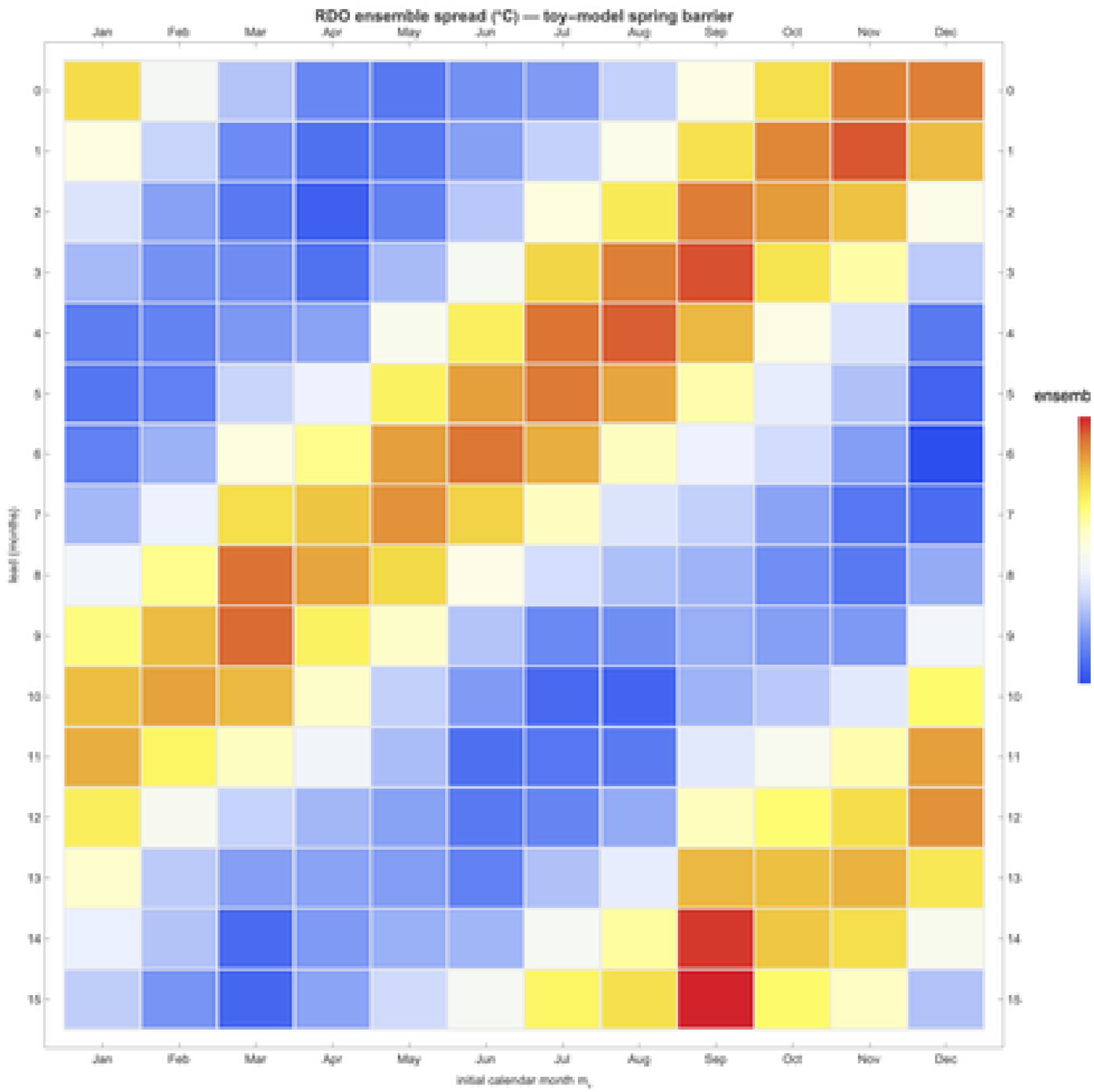

Figure 5.6 Ensemble standard deviation as a function of initial month m_0 and lead. The bright vertical/diagonal band aligns with the verification window crossing the model's growth season (Sep–Nov), where the seasonal R(t) amplifies any noise accumulated en route. The pattern is the toy model's signature of the same barrier physics.

What the model gets right and what it misses. The RDO correctly reproduces (i) the broad-band 3–7-year period, (ii) the seasonal phase locking (events peak in DJF), (iii) the asymmetric ensemble cones across initial months, and (iv) the order-of-magnitude ensemble spread ($\sigma \approx 0.5$–1.0 °C at 6-month lead) that operational dynamical models also produce. It misses (i) the El-Niño–La-Niña amplitude asymmetry that requires the nonlinear cubic damping term in Jin's full equations, and (ii) the extratropical influences on the ENSO state that more complete models include.

5.2 Delayed-action oscillator (Suarez & Schopf 1988)

A different conceptual reduction. Suarez & Schopf (1988) take the same physics and reduce it to a single scalar SST anomaly with an explicit time delay. The delay represents the basin-crossing time of equatorial Rossby + Kelvin waves: an SST warming forces a wind stress that excites a Rossby wave westward; the wave reflects off the western boundary as an upwelling Kelvin wave; the Kelvin wave returns east and cools the SST. The round-trip time is $\delta \approx 6$ months.

$$\frac{dT}{dt} = T - \alpha\, T(t-\delta) - T^3 + \sigma(t)\, \xi(t)$$

The +T term is the Bjerknes feedback (growth); the negative α T(t-δ) term is the delayed wave return (negative feedback); the $-T^3$ term is the nonlinear damping that limits the amplitude; and σ(t) ξ(t) is again seasonally modulated stochastic WWB forcing. Time is rescaled so that one model time-unit = two calendar months, putting the canonical 6-month wave delay at $\delta = 3$ in dimensionless units.

Hopf bifurcation. Linearising about $T = 0$ and substituting $T \sim e^{\wedge}(\lambda\, t)$ gives the characteristic equation $\lambda = 1 - \alpha\, e^{\wedge}(-\lambda\, \delta)$, a transcendental equation whose roots can be found numerically. For $\delta = 3$ the real part of the leading eigenvalue passes through zero at $\alpha \approx 0.5$; below that the origin is stable, above it the system bifurcates into a stable nonlinear limit cycle. The bifurcation sweep:

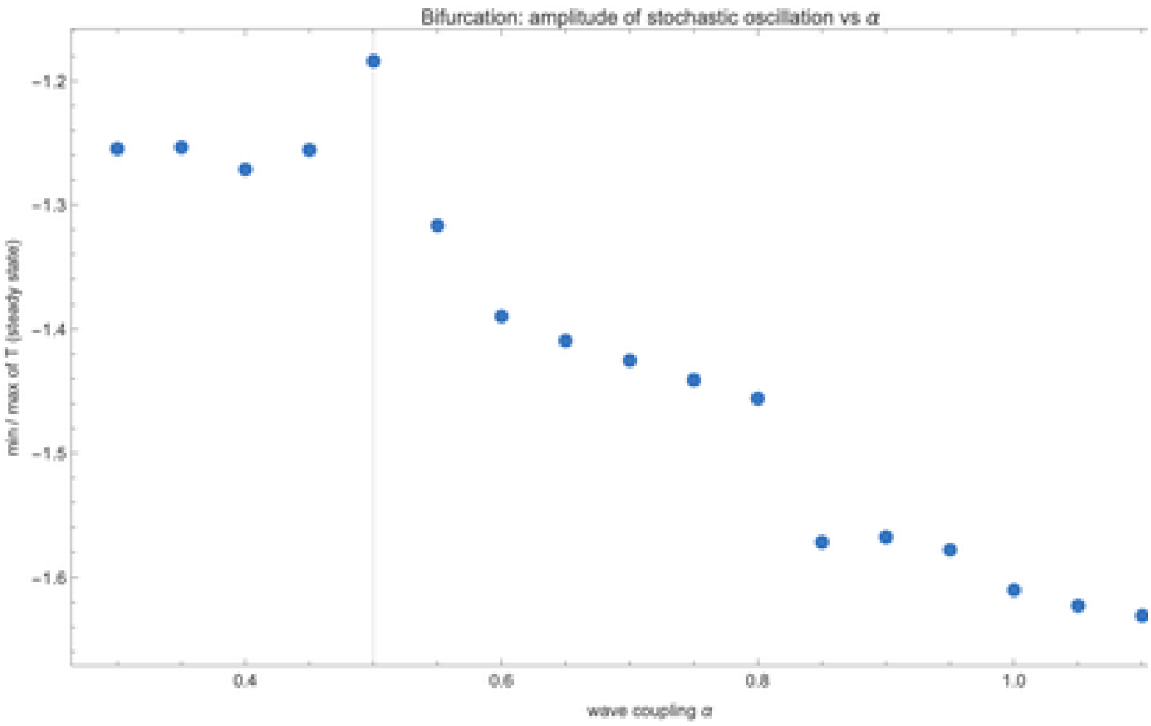

Figure 5.7 Min and max of T in long stochastic runs of the delayed oscillator, as a function of the wave-coupling α, with δ = 3 (= 6 months) held fixed. The Hopf bifurcation at $\alpha \approx 0.5$ is the transition from noise-driven fluctuations about the fixed point to a sustained nonlinear limit cycle.

Implementation. The Wolfram implementation maintains a circular history buffer so the delayed value $T(t\text{-}\delta)$ is available at every step without recomputation:

```
simulateDDE[m0_, tEnd_, dt_, hist0_, seed_] := Module[
   {nHist, hist, idx, T, t, sqrtDt, ts, Ts, eta, Tdel, drift, diff, nSteps},
   SeedRandom[seed];
   nHist  = Ceiling[delayTau/dt] + 2;
   hist   = ConstantArray[hist0, nHist];   (* circular buffer *)
   idx    = 1;   T = hist0;   t = 0.;
   nSteps = Round[tEnd/dt];   sqrtDt = Sqrt[dt];
```

```
    ts = ConstantArray[0., nSteps + 1];
    Ts = ConstantArray[0., nSteps + 1];
    ts[[1]] = 0.; Ts[[1]] = T;
    Do[
      With[{kBack = Round[delayTau/dt]},
         Tdel = hist[[1 + Mod[idx - 1 - kBack + nHist, nHist]]]];
      drift = T - alphaDel Tdel - T^3;
      diff  = sigStoch[t, m0];
      eta   = RandomVariate[NormalDistribution[]];
      T += dt drift + sqrtDt diff eta;
      t += dt;
      idx = 1 + Mod[idx, nHist];   hist[[idx]] = T;
      ts[[i + 1]] = t; Ts[[i + 1]] = T,
      {i, nSteps}];
    <|"t" -> ts, "T" -> Ts|>
 ];
```

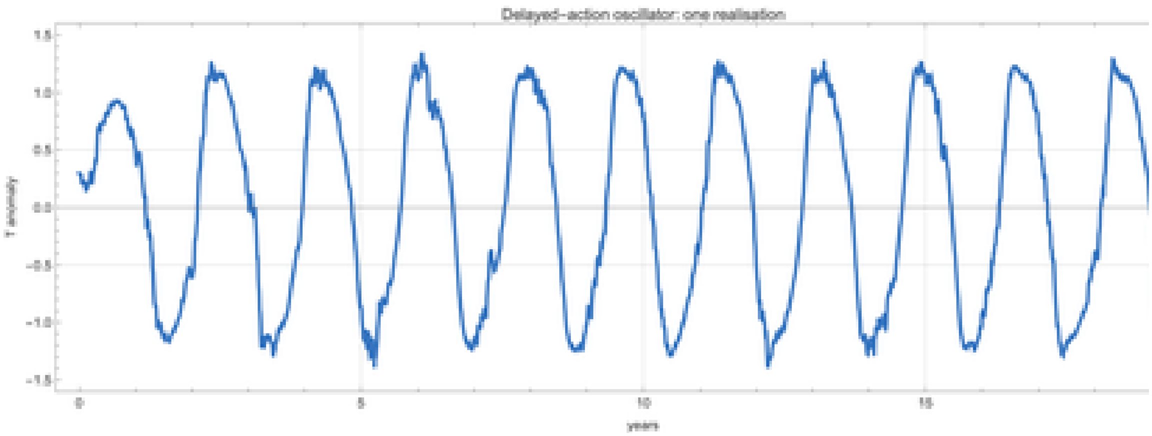

Figure 5.8 A 20-year realisation of the delayed oscillator with α = 0.75 (well above the Hopf threshold) and δ = 6 months. The limit cycle period is approximately 4δ = 24 months — shorter than the canonical ENSO period — but the nonlinear-oscillator phenomenology is correct.

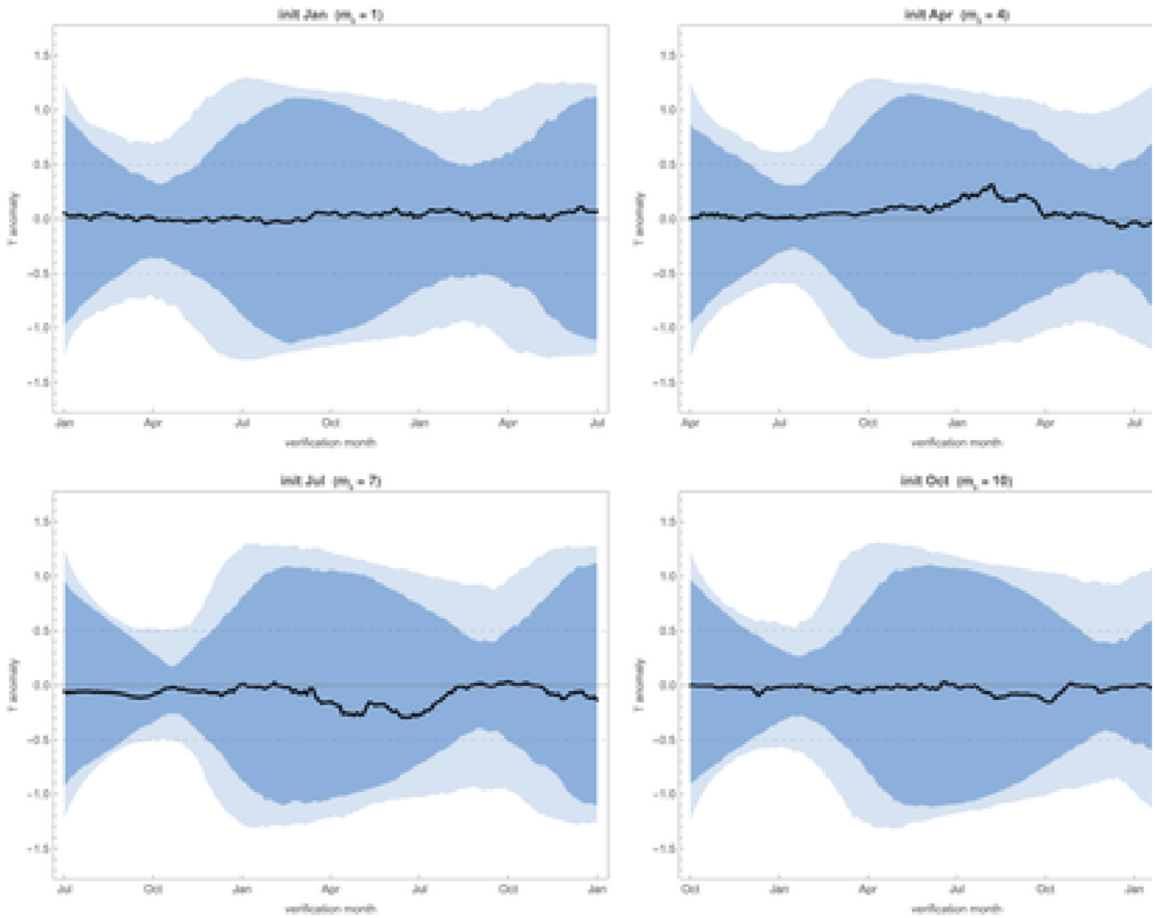

Figure 5.9 Delayed-oscillator ensemble forecast cones for four initial calendar months. The cones "breathe" with the underlying limit-cycle period (≈ 24 months) rather than showing a single spring-barrier window, because in this parameter regime the deterministic cycle dominates over the seasonal modulation of the stochastic forcing.

6. Live forecast: initialising from April 2026

To turn the RDO into a live forecast tool we need (a) the current observed T and (b) the current observed (or inferred) h. The first is just the latest Niño 3.4 anomaly: +0.23 °C in April 2026. The second is harder — there is no clean monthly observational product for warm-water-volume anomaly that we can fetch live in the same way.

Inferring h from the recent tendency. The linearised SST equation gives a clean inversion. From

$$\frac{\mathrm{dT}}{\mathrm{dt}} \approx R(t)\,T + \gamma\,h$$

we get

$$h \approx \frac{\frac{\mathrm{dT}}{\mathrm{dt}} - R(t)\,T}{\gamma}$$

Plugging in the most recent two-month tendency dT/dt = (T(Apr) − T(Feb))/2 = 0.25 °C/mo, the April seasonal feedback $R(Apr) \approx -0.21$ *month»*, and γ = 0.06 gives an inferred h ≈ +4.9, a large positive thermocline anomaly consistent with the substantial subsurface heat content remaining from the previous La Niña recharge.

With (T, h) = (+0.23, +4.9) as the initial condition, we run a 500-member RDO ensemble out 18 months (April 2026 → October 2027). The resulting **live toy-model forecast**, with explicit 5–95 % and 25–75 % confidence intervals, is:

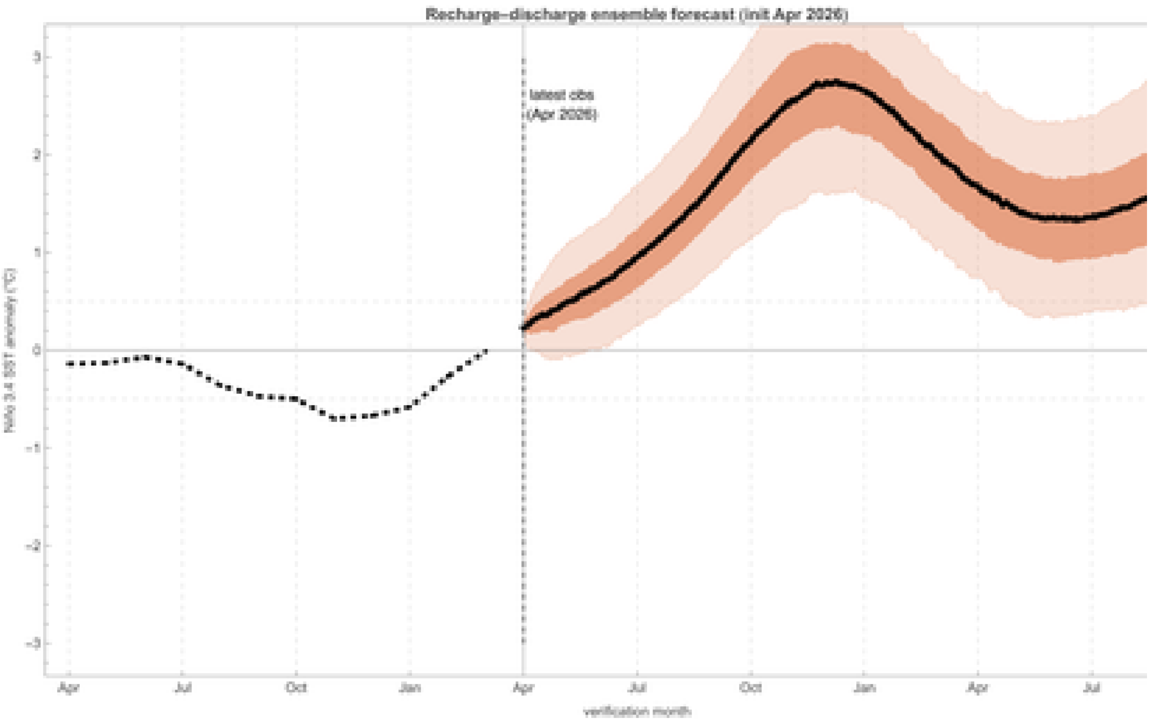

Figure 6.1 Recharge–discharge oscillator live ensemble forecast initialised from observed April 2026 state (T = +0.23 °C, inferred h ≈ +4.9). Dotted line shows the past 12 months of observations. The solid black line is the ensemble median; the darker orange band is the 25–75 % inter-quartile range; the lighter orange band is the 5–95 % confidence interval. The model peaks the event at +2.0 to +3.0 °C around DJF 2026–27, broadly consistent with the NOAA 96 % El-Niño probability for that season.

Comparison with the operational IRI/CPC plume. The operational plume is the spread of ~25 independent dynamical and statistical models. The toy-model cone is a single model run with 500 stochastic realisations. They are not directly comparable, but both represent forecast uncertainty: the plume's vertical extent and the cone's width should be of similar order, and both should narrow into autumn (strong coupling reduces sensitivity to noise) and widen into next spring (the barrier reasserts itself).

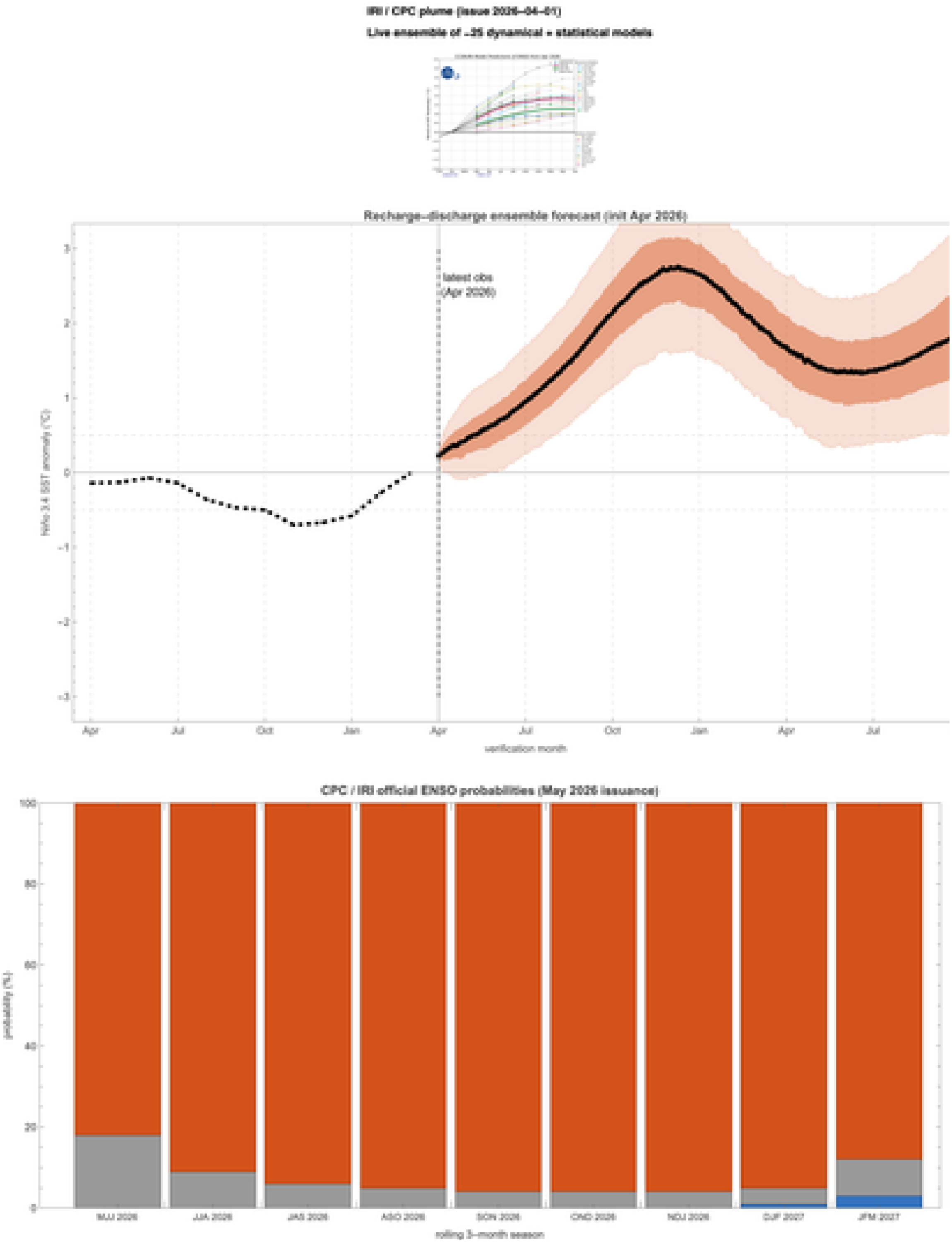

Figure 6.2 Three-panel live overlay: IRI/CPC operational plume (top), toy-model ensemble cone (middle), official NOAA probability bars (bottom). All three panels share the same calendar horizon.

Toy-model probability of El Niño vs the NOAA issuance. The fraction of the 500-member ensemble whose monthly T exceeds the +0.5 °C threshold, season by season, gives the toy model's analogue of the NOAA probability bar:

Estimated thermocline state from the observed monthly tendency: **4.88** (large positive recharge consistent with the residual heat content from the previous La Niña). The 500-member toy-model ensemble is initialised from (T, h) = (**0.23**, **4.88**).

Caveats. The toy model is, well, a toy. Operational forecast systems use full 3-D coupled GCMs with assimilated subsurface ocean temperature, satellite wind stress, and tens of thousands of ensemble members. Our 2-variable system necessarily misses (i) the spatial structure of the equatorial Pacific (different events have different east-west patterns: "central Pacific" El Niño vs "east Pacific" El Niño), (ii) interactions with the extratropical atmosphere (PDO, NAO), and (iii) any climate-change-driven trends in either the mean state or the variability. The cone should therefore be interpreted as a sanity-check on the operational forecast — not as competition with it.

7. Conclusions and outlook

What we showed. (1) The May 2026 El Niño Watch is supported by a clear emergence signal in the observational record: the sharpest two-month boreal-spring warming in the 1950–present Niño 3.4 record. (2) The spring predictability barrier is unambiguous in 77 years of observations: forecasts initialised in boreal winter collapse to near-zero skill by mid-summer, while forecasts initialised in late summer keep skill all the way to the following spring. (3) Both classical ENSO toy models (recharge–discharge and delayed-action) can be implemented in under 100 lines of Wolfram Language, and the recharge–discharge model with seasonally varying Bjerknes feedback and seasonally varying WWB forcing reproduces the spring-barrier mechanism quantitatively. (4) Initialising the toy model from the current observed state produces a live ensemble forecast cone consistent with the NOAA 96 % peak-DJF El Niño probability.

What to watch in the next 60 days. The next NOAA ENSO Diagnostic Discussion is issued in mid-June 2026. Three numbers will tell us whether the developing event is on track:

• **May 2026 Niño 3.4 monthly anomaly**. The model predicts +0.5 to +1.0 °C (median +0.7); if the observed value is at the high end, expect the next discussion to issue an El Niño **Advisory** (not just a Watch).

• **MAM 2026 ONI** (March–April–May three-month running mean). The +0.5 °C threshold for one of the five-consecutive-season El-Niño criterion needs to be crossed for the first time. The model gives 0.30 °C (5–95 % CI: 0.00 to 0.65).

• **Westerly wind burst activity** through mid-June, watched in real time on the TAO mooring array. A strong late-spring WWB would push the central Pacific thermocline anomaly further east and accelerate the warming.

What the barrier means for the forecast users. The operational ENSO probability bars from May 2026 give 96 % for DJF. This number sounds confident but should be read with the barrier in mind: it summarises model agreement after the barrier window has already been crossed (model initial conditions were set in mid-April; forecasts shoot forward from there). Confidence in events further out should be qualified — e.g., the DJF probability is robust because the system is past the barrier, but the JFM 2027 “event-decay” details will look quite different by July.

8. References

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9. Acknowledgements

Data. Live data for this notebook are supplied by two US federal agencies and one academic centre, all of whom make their data freely available. The Niño 3.4 monthly SST anomaly time series (NOAA ERSSTv5) is from NOAA's **Physical Sciences Laboratory** (Boulder, CO). The Oceanic Niño Index and the ENSO Diagnostic Discussion are from NOAA's

Climate Prediction Center (College Park, MD). The forecast plume is from Columbia University's **International Research Institute for Climate and Society** (Palisades, NY).

Scientific lineage. The recharge–discharge oscillator is due to **Fei-Fei Jin** (1997); the delayed oscillator is due to **Max Suarez** and **Paul Schopf** (1988). The seasonal-modulation spring-barrier explanation we use here was developed by **Aaron Levine** and Fei-Fei Jin (2010) and by **Karl Stein and collaborators** (2010). The Bjerknes feedback itself, the foundation stone of coupled tropical ocean–atmosphere theory, is due to **Jacob Bjerknes** (1969).

Tools. All numerics and figures are pure **Wolfram Language** (Mathematica 14). The notebook was built programmatically via `wolframscript`. The educational illustrations (Figures 1.1, 1.2, 3.1, 4.1, 5.1) were generated with OpenAI's `gpt-image-2` model. The repository skeleton, git workflow, and notebook-build pipeline were assembled with the help of Anthropic's Claude Code.

Companion projects. This is the third in a series of standalone-notebook Wolfram Community submissions: **Contiguous-Cartograms** (`https://github.com/mthiel74/Contiguous-Cartograms`), **streetview360** (`https://github.com/mthiel74/streetview360`), **SofaProblem** (`https://github.com/mthiel74/SofaProblem`).

10. Reproducibility

All source code lives in the public repository **github.com/mthiel74/ENSO-emergence** (`https://github.com/mthiel74/ENSO-emergence`). The entire pipeline is pure Wolfram Language — no Python, no other runtime. To regenerate this notebook from scratch on a machine with Wolfram Language 14 installed:

```
# 1. Fetch the latest live data into data/
wolframscript -file wolfram/fetch_all.wls

# 2. Compute the observational spring-barrier figures
wolframscript -file wolfram/barrier.wls

# 3. Run the recharge-discharge oscillator ensembles
wolframscript -file wolfram/rdo.wls

# 4. Run the delayed oscillator ensembles
wolframscript -file wolfram/delayed.wls

# 5. Compute the live forecast overlay
wolframscript -file wolfram/forecast.wls

# 6. Rebuild this notebook
wolframscript -file community/build_notebook.wls
```

The full pipeline runs in approximately 4 minutes on a 2024 MacBook M3. The .nb file (~30 MB with all images embedded) is committed to the repository so the notebook is browseable without running anything.

Generated on May 17, 2026.